

AVEVA Academic Competition 2026 – Part 2

Scope of Work – Part 2

In Part 1 of the AVEVA Academic Competition, you focused on training a Machine Learning (ML) model to predict performance of a cumene process based on a set of inputs from the plant. You also built an interactive dashboard for use by operators and executives at your company to monitor the process.

In Part 2, you will integrate your ML model into AVEVA Process Simulation to create a high-fidelity model that simulates the plant's behavior. A simplified PFD for the process is shown in **Figure 1** below. Using the ML model you trained in Part 1:

1. **Build a simulation model** of the cumene process using AVEVA Process Simulation. Your simulation model should include all equipment shown in **Figure 1**, except any model(s) you have chosen to represent with your ML model. Additional information required to build the model is given in **Appendix A – Simulation Inputs**.

NOTE: if a specific model specification is not provided in **Appendix A – Simulation Inputs**, use the default specification from AVEVA Process Simulation. You can choose to update the specification based on design heuristics, but this is not required.

2. **Integrate your trained ML model** into the simulation. To help you get started, we have included an example in **Appendix B – Incorporating an ONNX model into AVEVA Process Simulation**.

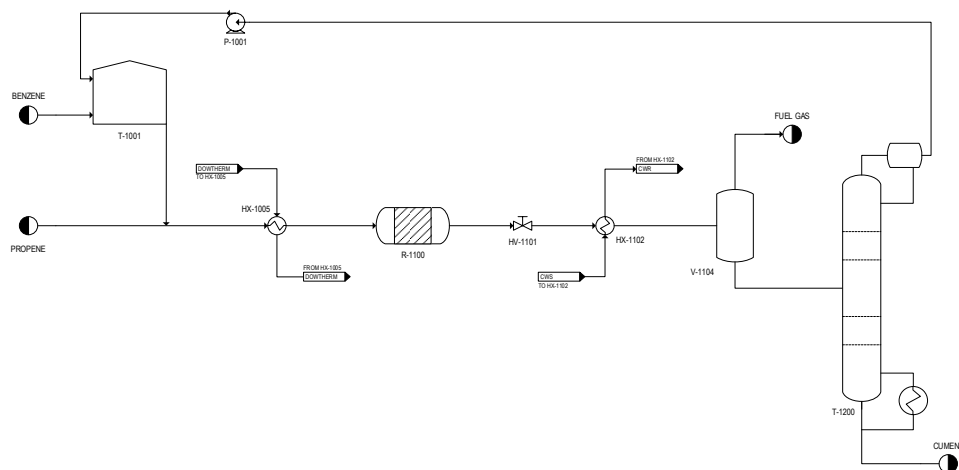


Figure 1 - Simplified PFD for AVEVA Process Simulation model

Deliverables – Part 2

In Part 2, you will decide how to present your final model. Either:

Prepare a **report** (no more than 3 pages) or

Prepare and record a **PowerPoint** presentation (no more than 8 minutes)

Regardless of what option you choose, your final deliverable should include:

1. A short description/discussion of the simulation model you have built. Include at least one screenshot showing an overview of your simulation model.
2. A description/discussion of how you have integrated your ML model into the simulation. Be sure to describe the inputs and outputs of the ML model and how the model was integrated into the simulation. If you have replaced equipment items from the PFD with your ML model, be sure to describe why you replaced these models.

HINT: in Part 1 of the competition, you were also asked to describe the inputs and outputs of your ML model. Your report from Part 1 should focus on the available data and how that was used to train the model. In Part 2, focus on how these translate to specific input and output variables in the simulation.

3. Case study results showing the impact of 1) reactor temperature, 2) feed flow rate, and 3) benzene to propene ratio on the cumene production. Use the following bounds for the case study:
 - Reactor temperature from 335 °C – 365 °C
 - Reactor feed flow rate from 15,000 – 18,000 kg/h
 - Benzene to propene ratio from 4 – 6

At a minimum, your case study results should include the molar concentration of cumene at the outlet of the reactor (if available through your model) and the mass flow rate of the cumene product. Other relevant information that describes the impact of these changes on the process will also be accepted. Be sure to include figures of the case study results, a discussion for each scenario, and whether you believe the results to be accurate (e.g., if you were an engineer, how would you convince others that your model is producing the correct results?).

4. A short description/discussion of the advantages and disadvantages of your hybrid model compared to a standalone ML model or process simulation model.

If you have chosen to present your findings in a report, your report should be prepared using the following format:

- **Paper Size:** Letter (8.5 x 11in) or A4 (210 x 297mm)
- **Font Size:** 12
- **Font:** Aptos, Arial, Calibri, or Times New Roman
- **Line Spacing:** 1.15
- **Margins:** 1" (please include your last name in the header of the first page)
- **File Name and Format:** [LAST NAME]_2026AcademicCompetition_Part2.pdf

If you have chosen the PowerPoint option, you will need to provide a video of your presentation with narration. You can choose to record yourself presenting or simply provide background audio. Your final video can be sent as a file or uploaded to YouTube for our judges to access. In case of technical difficulties, please also export your PowerPoint presentation as a PDF using the file name and format above. Please include your name on the title slide of the presentation.

Grading Rubric and Submission Instructions – Part 2

Grading Rubric

Your submission will be graded according to the following rubric. Use the descriptions provided in the section above (Deliverables – Part 2) for more guidance.

Deliverable	Weight
Simulation Model Points will be awarded for successfully creating the simulation model shown in Figure 1 .	15%
Integration of Machine Learning model Appropriate justification and explanation should be provided describing how you integrated your machine learning model into the simulation.	25%
Case study results Case study results will be judged based on the clarity of provided data and relevance of the accompanying discussion.	25%
Hybrid modeling discussion Points will be awarded for identifying appropriate advantages and disadvantages of hybrid modeling.	15%
Formatting and presentation	20%

Submission Instructions

You will need to submit all deliverables for the competition by **Saturday, February 28, 2026** at **11:59pm PST**. Be sure to include:

1. Part 1 report
2. Part 2 report or video presentation and backup PDF of your PowerPoint slides
3. Supplemental video showcasing your dashboards from Part 1 (*Optional*)

Please email all deliverables for your submission to academic.program@aveva.com. If your video files are too large to attach, please upload elsewhere and provide a link to access.

Appendix A – Simulation Inputs

Process Description (copied from Part 1)

NOTE: the process description and PFDs provided below have been copied from Part 1 of the competition. The PFDs shown in **Figure 2** and **Figure 3** are more detailed than are required for your simulation model. Refer to **Figure 1** instead.

In addition to the information provided below, assume that the total mass flow rate of the benzene feed (FT-1003) is 15,000 kg/h under normal operating conditions.

Process Flow Diagrams (PFDs) for the cumene process are shown in **Figure 2** and **Figure 3** below. Benzene is combined with propene in a 5:1 ratio of benzene to propene. In HX-1005, the mixed feed is heated to 350°C using a DOWTHERM™ heat transfer utility. The hot feed stream then enters an adiabatic packed bed reactor (R-1100) to produce the cumene product. The reactor effluent is cooled to 90°C in HX-1102 before entering drum V-1104 to vent off light gases, which are used as fuel gas in other areas of the plant. The remaining benzene/cumene mixture is fed to a 13-stage distillation column (T-1200) where a 99 mol% cumene product is recovered in the bottom product stream. From the top of the column, benzene (containing ≤ 1 mol% cumene) is recycled back to the front of the process and combined with fresh benzene in feed tank T-1001.

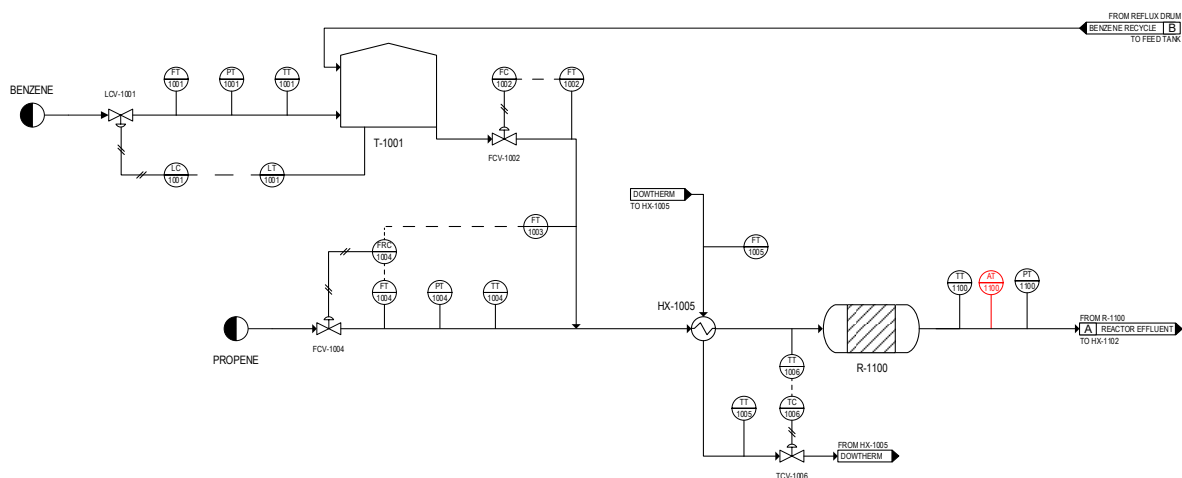


Figure 2 - Detailed process flow diagram of reactor section (copied from Part 1)

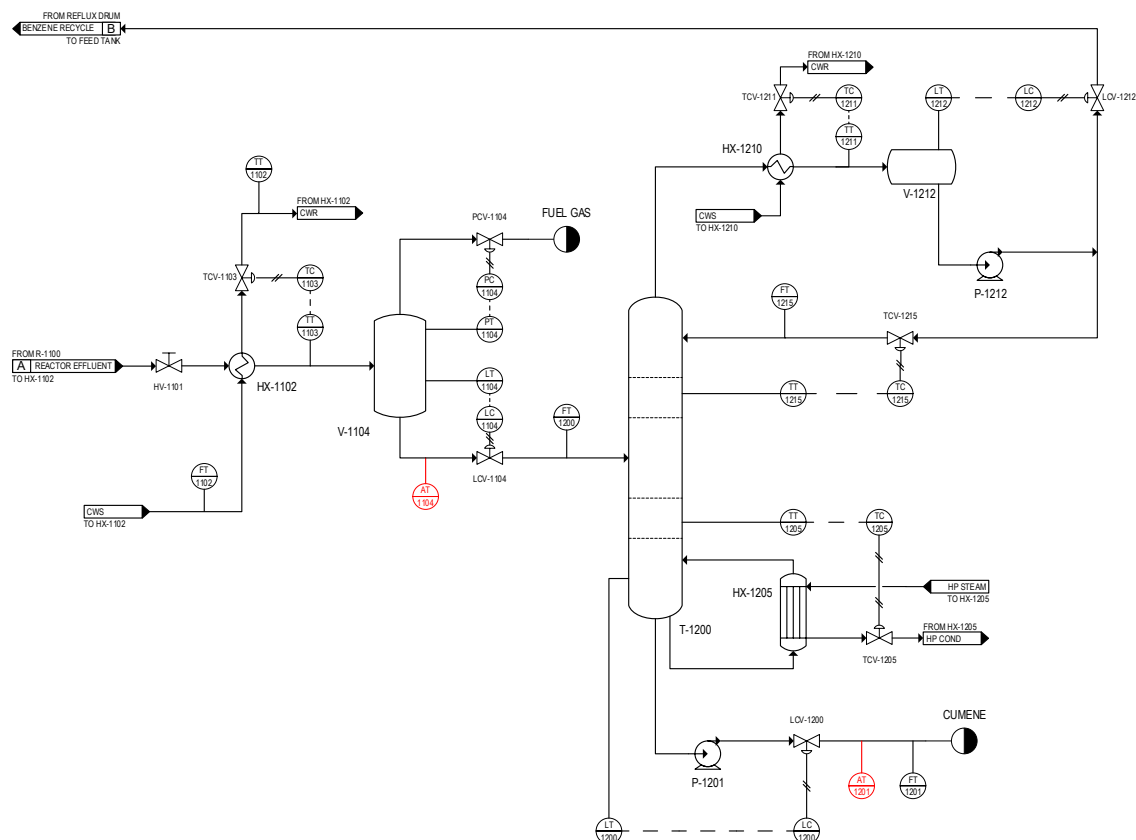


Figure 3 - Process Flow Diagram (PFD) for the separation section of the cumene process

Simulation Specifications

Table 1 – Thermodynamic properties for fluid

Variable / Parameter	Value
Thermo Type	Compositional
System	Soave-Redlich-Kwong (SRK)
Phases	Vapor/Liquid (VLE)

Table 2 - Components for fluid

Name	Full Name	Formula	Composition
BENZENE	Benzene	C6H6	0.63
PROPENE	Propylene	C3H6	0.34
PROPANE	Propane	C3H8	0.02
CUMENE	Cumene	C9H12	0.01

Table 3 – Source specifications

Model	Property	Variable	Value
PROPENE	Pressure	P	31 bar
	Temperature	T	210 °C
	Molar Composition	M[BENZENE]	0 kmol
		M[PROPENE]	0.95 kmol
		M[PROPANE]	0.05 kmol
		M[CUMENE]	0 kmol
BENZENE	Pressure	P	32 bar
	Temperature	T	30 °C
	Molar Composition	M[BENZENE]	1 kmol
		M[PROPENE]	0 kmol
		M[PROPANE]	0 kmol
		M[CUMENE]	0 kmol

Table 4 – Heat exchanger specifications

Model	Property	Variable	Value
HX-1005	Overall heat transfer coefficient	U	568 W/m ² -K
	Tube Outlet Temperature	Tto	350 °C
	Shell Outlet Temperature	Tso	350 °C
	Shell Side Pressure Drop	DPs	0.25 bar
	Tube Side Pressure Drop	DPt	0.25 bar
HX-1102	Overall heat transfer coefficient	U	450 W/m ² -K
	Tube Outlet Temperature	Tto	40 °C
	Shell Outlet Temperature	Tso	90 °C
	Shell Side Pressure Drop	DPs	0.25 bar
	Tube Side Pressure Drop	DPt	0.25 bar

Table 5 – Reactor specifications

Model	Property	Variable	Value
R-1100	Thermal operation	Operation	Adiabatic
	Reaction kinetics	RxnModelType	Unknown
	Reaction rate basis	Basis	CatalystMass
	Number of tubes	Nt	234
	Tube length	L	2 m
	Tube diameter	D	76.2 mm
	Catalyst density	rhoc	1600 kg/m ³

Table 6 – Column specifications

Model	Property	Variable	Value
T-1200	Number of ideal stages	NStages	13
	Feed stage	FeedStage	7
	Condenser option	Condenser	Internal*
	Reboiler option	Reboiler	Internal*
	Sump option	Sump	Single
	Top stage pressure	Ptop	1.75 bar
	Stage pressure drop	DPstage	0.01 bar
	Condenser pressure drop	DPcond	0.25 bar
	Reboiler pressure drop	DPreb	0.25 bar
	Condenser vapor fraction	VFcond	0 mol frac

Table 7 – Valve specifications

Model	Property	Variable	Value
HV-1101	Outlet pressure	P2	1.8 bar

Table 8 – Pump specifications

Model	Property	Variable	Value
P-1001	Outlet pressure	P2	32 bar

Table 9 – Available utilities for process

Utility	Property	Variable	Value
Dowtherm	Pressure	P	5 bar
	Temperature	T	550 °C
Cooling Water	Pressure	P	5 bar
	Temperature	T	30 °C

Appendix B – Incorporating an ONNX model into AVEVA Process Simulation

Note: we have prepared the example below to help you integrate your ONNX model into AVEVA Process Simulation. The inputs and outputs are different than the model you have trained for the competition.

Imagine you are trying to model the behavior of a pump in your process but do not have access to the performance curve. In this example, we will show how you can use an ML model in AVEVA Process Simulation to predict the **head** and **efficiency** of the pump.

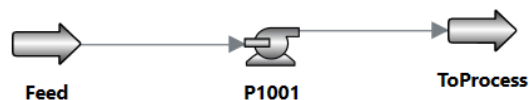


Figure 4 - A simple pump model in AVEVA Process Simulation

Step 1: Determine inputs and outputs

In this example, we use **volumetric flow rate (P1001.Q)** and **motor speed (P1001.Speed)** as the input variables for the ML model. These values are easy to measure in the process and describe the behavior of a pump well (to convince yourself of this, look at a standard performance curve for a pump).

We have chosen **discharge pressure (P1001.P2)** and **power (P1001.Power)** as output variables. These are easy to measure in the process and will be used to calculate the corresponding **head** and **efficiency** in AVEVA Process Simulation.

Step 2: Train your ML model

In this example, we have chosen a k-nearest neighbors regression model from scikit-learn to estimate the discharge pressure and power. A single line of the code is provided below to demonstrate the output of the model. Remember this is only an example.

```
KNNmodel = KNeighborsRegressor(n_neighbors=2).fit(X_train, y_train)
```

Step 3: Export as an ONNX model

After training the ML model, we need to export it to the ONNX format for use within AVEVA Process Simulation. The Python code below converts the KNNmodel from scikit-learn to an equivalent model in ONNX format.

NOTE: we recommend converting your dataset to a **float32** variable type for compatibility with AVEVA Process Simulation.

```
from skl2onnx import to_onnx
onnx = to_onnx(KNNmodel, X_train.to_numpy(dtype="float32"))

with open("PumpModelONNX.onnx", "wb") as f:
    f.write(onnx.SerializeToString())
```

Step 4: Import the Preview library in AVEVA Process Simulation

The Preview library of AVEVA Process Simulation includes the **ONNXAdaptor** model that we will use to call our trained ML model. To import the Preview library, first check that you are using the **Model Writing** role. Then, click the [+] button in the bottom left corner of the screen (under the list of libraries), select **Import Library**, and open the **Preview** library (...\\SimCentral Examples\\Libraries\\Preview.libx).

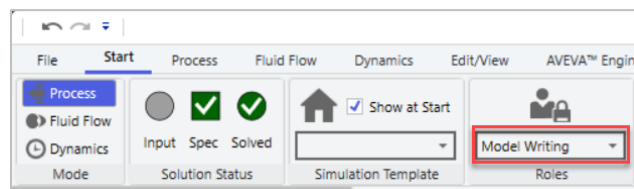


Figure 5 - Model Writing role in AVEVA Process Simulation

Step 5: Configure the ONNX adaptor

To start, we drag/drop the **ONNXAdaptor** model into the simulation to create model **ONNX1**.

From the Mini Inspector, update the **Path** parameter to direct the adapter to the ONNX model you created in Step 3 above. To populate the **Description** field, change the **Setup** parameter to **Solve** then change back to **Configure**. Copy the information from the **Description** parameter to **NI**, **NO**, **InputLabel**, and **OutputLabel** as shown below.

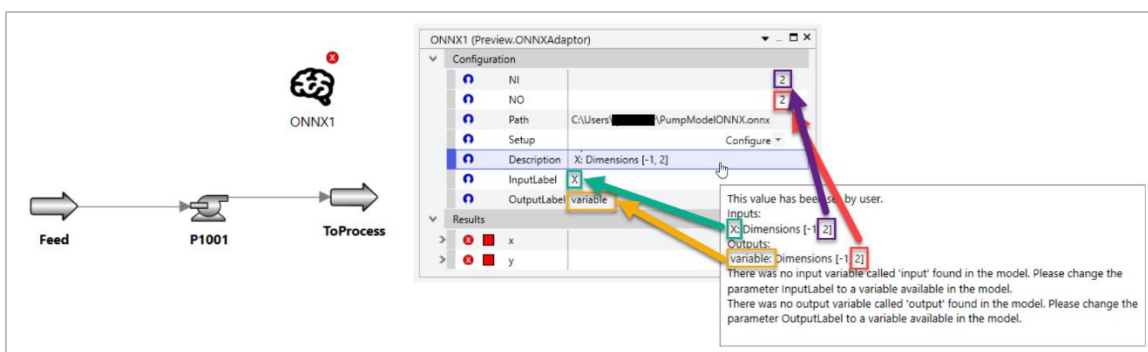


Figure 6 - Initial configuration of ONNX1 model

Step 6: Connect the ONNX model inputs to variables in the simulation

Next, we will indicate which variables in the simulation correspond to the inputs of the ML model. In this example, we will use flowsheet equations. The model help file for the **ONNXAdaptor** explains how you can embed this model within another model if desired.

Recall that we identified **P1001.Speed** and **P1001.Q** as inputs in Step 1. We add simple flowsheet equations setting these equal to the ONNX input variables:

$$\text{Eqn1: } \text{ONNX1.x}[1] = \text{P1001.Speed}$$

$$\text{Eqn2: } \text{ONNX1.x}[2] = \text{P1001.Q} * 3600$$

NOTE: be careful of units! Double check that the values displayed in the ONNXAdaptor model match the values and units of measure you used to train your model. Equations in AVEVA Process Simulation use the internal, default UOMs which may be different than the ones you have used to train your model.

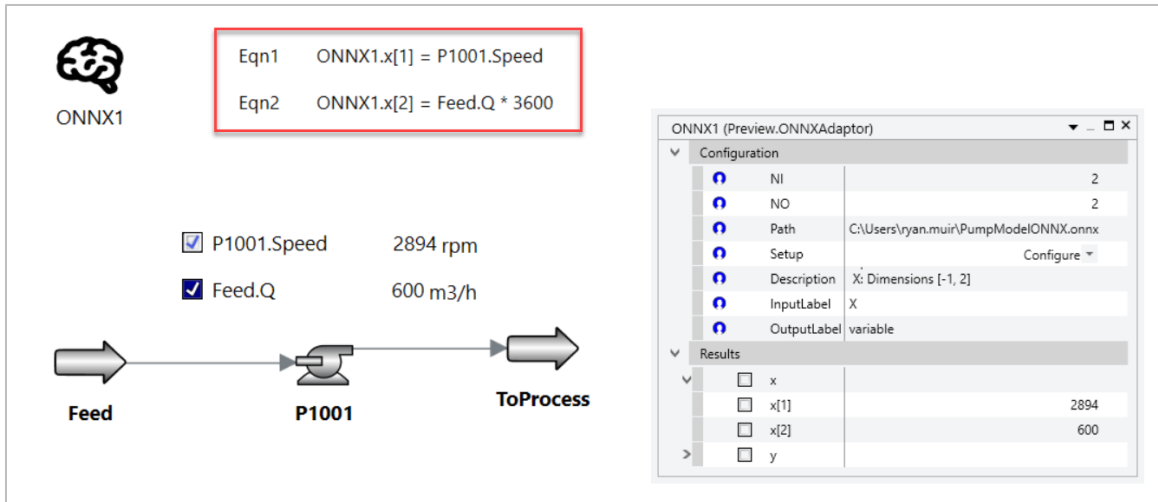


Figure 7 - Flowsheet equations defining the inputs to the ONNX model

Step 7: Confirm the ONNX model is working as expected

After configuring the model inputs, change the **Setup** parameter to **Solve**. Check that the ONNX model is producing outputs in the **Results** group. If not, check your log file for more information (...\\SimCentral\\Logs\\Server.log).

Step 8: Map the outputs to variables in the simulation

After the ONNX model has been solved, we can map the output values to their corresponding variables in the simulation. In this example, we will use flowsheet equations to complete the mapping.

Earlier, we identified the outputs of the ML model as **P1001.P2** and **P1001.Power**, so we add equations mapping these variables to the outputs of the ONNX model:

$$\text{Eqn3: } \text{P1001.P2} = \text{ONNX1.y}[1]$$

$$\text{Eqn4: } \text{P1001.Power} = \text{ONNX1.y}[2]$$

After making these specifications, the model is overspecified by two variables. Therefore, we must choose two variables to unspecify. **P1001.P2** and **P1001.Power** would be the obvious choices, but neither is specified in the model by default. Instead, we unspecify **P1001.DP** and **P1001.eta** to re-square the model. These are in the same equation sets as **P1001.P2** and **P1001.Power**, respectively. The model is now complete.

Understanding which variables to unspecify is an important part of configuring your ONNX model. Your selections (and the corresponding justification of those selections) will be used when grading your responses for the competition.

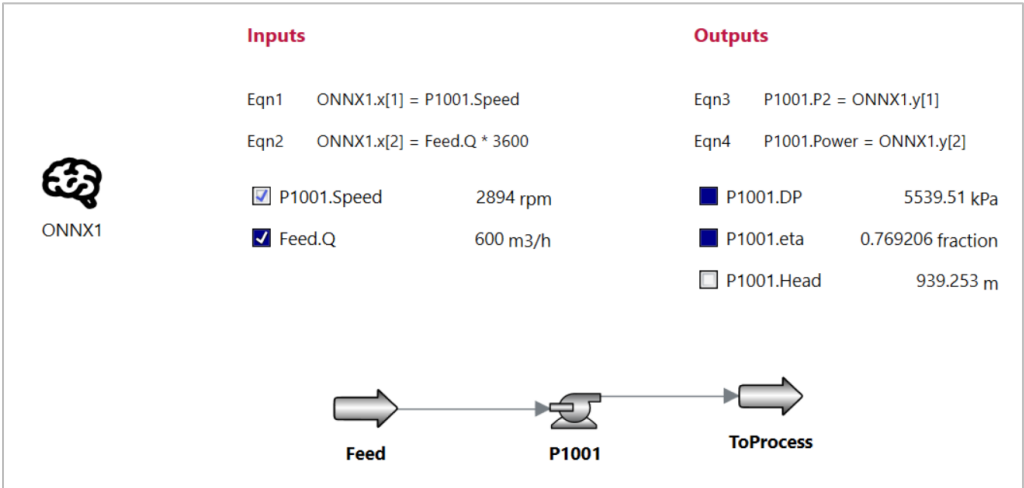


Figure 8 - Final ONNX model to estimate pump performance from input data